## Sprint 3-  Project Manager Quality Assurance Review

**Sprint**: Sprint 3 (October 27 - November 9, 2026)
**QA Reviewer**: Benjamin Popso
**File**: packet-tracer/US-23 Access-Layer Switching Protection.pkt

### QA Testing Summary

A final quality assurance review was performed on the network to evaluate Sprint 3 implementation and confirm that newly integrated features coexist with the established network infrastructure before Sprint 4.

**Verified results**:

- **VLANs, Routing, and OSPF- PASS**: Verified expected VLAN configuration, OSPF adjacencies, routing tables, default routes, and trunk configurations across both multilayer switches.
- **Spanning Tree Protocol- PASS**: Confirmed the intended root bridge assignments for VLANs 10, 20, 30, 40, 50, and 99. Verified distribution-layer forwarding and representative access-switch port states.
- **Extended ACLs- PASS**: Tested permitted and prohibited traffic from Finance, General Staff, and Wireless VLANs. Authorized server communication succeeded, while restricted destinations were denied.
- **DHCP/DNS Integration-  PASS**: Successfully renewed DHCP leases from Finance and General Staff clients and verified internal DNS resolution of `resource.kfp.internal` to `10.20.40.13`.
- **Unused-Port Protection-  PASS**: Confirmed designated unused ports across SW1–SW4 are assigned to VLAN 999 and administratively shut down, without disrupting active infrastructure.
- **Port Security- PASS**: Verified maximum-one-MAC enforcement and sticky MAC learning on designated PC-facing interfaces. A controlled unauthorized MAC test triggered secure shutdown and err-disable, followed by successful recovery and restored connectivity.
- **PortFast/BPDU Guard- PASS**: Verified appropriate configuration on endpoint-facing interfaces, operational PortFast behavior, and functioning uplink connections.
- **NAT/PAT- PASS** (previously tested): NAT/PAT was tested separately, with implementation and peer-verification evidence maintained in the project repository.
- **Wireless Security and Connectivity- PASS**: WPA2-Enterprise authentication, wireless DHCP, VLAN 50 segregation, local switching, and permitted/denied connectivity were demonstrated during implementation and independent peer testing.

- **NTP/Syslog- PASS:** Verified centralized Syslog collection of timestamped events from multiple network devices and successful NTP synchronization.

**Observations and Future Improvements**

- SW1 was observed acting as the root bridge for VLAN 999 (unused/native VLAN). While this does not affect the intended root selection for production VLANs, the team may standardize VLAN 999 root bridge placement during a later hardening sprint.

Additional potential improvements include evaluating port-security protections for dedicated server interfaces, subject to operational requirements.

**QA Conclusion**

- Sprint 3 integration tasks and acceptance tests were reviewed and tested successfully. The network is ready to advance into Sprint 4 for comprehensive regression testing, failure-recovery validation, further infrastructure hardening, and preparation for the November 20 in-person MVP demonstration.